

Rising Waters: A Machine Learning Approach to Flood Prediction Solution Requirements

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Admin login for authorized access	High
2	Authorization levels	Only authorized users can predict floods	High
3	External interfaces	Uses Kaggle dataset	Medium
4	Prediction	Predict flood possibility	High
5	Reporting	Display prediction result	High
6	Business rules, etc.	Use trained ML model	High
7	Deployment	IBM Cloud deployment	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast prediction	< 2 seconds
2	Scalability	Cloud deployment	Supports multiple users
3	Security & Data Privacy	Secure application	Authorized access
4	Reliability & Availability	Accurate predictions	96.55% Accuracy
5	Usability & Accessibility	Easy interface	Beginner Friendly
6	Maintainability	Easy updates	Modular Code